

56.5 lm W⁻¹ and 26.3%, respectively, with CRI of 86. Even at 10000 and 15000 cd m⁻², high PEs (45.8 and 38.5 lm W⁻¹) and EQEs (24.3 and 22.1%) are maintained. Lighting sources are generally characterized by their total emitted light, which is a factor of 1.7-2.3 times more than the forward viewing efficiencies.^[1a] Therefore, the present device should have total PE and EQE values approaching 100 lm W⁻¹ and 50%, respectively, at 5000 cd m⁻², which are also the highest levels reported for WOLEDs with extra out-coupling and/or multi-layer tandem structures,^[14] as well as comparable to the most efficient lighting technologies.^[15]

Table 1. Summary of EL performance for Devices 1, 2, 3 and 4.

Device	V _{on} ^{a)} [V]	V ^{b)} [V]	η _p ^{b)} [lm W ⁻¹]	η _{ext} ^{b)} [%]	CIE _{x,y} ^{b)}
Device 1	2.5	3.8, 4.9, 5.6	74.5, 56.5, 45.8	27.3, 26.3, 24.3	(0.43, 0.46), (0.42, 0.45), (0.41, 0.45)
Device 2	2.4	4.0, 5.2, 6.1	32.4, 21.2, 16.0	15.5, 13.1, 11.5	(0.24, 0.48), (0.23, 0.42), (0.22, 0.36)
Device 3	2.6	4.7, 6.1, 6.9	42.8, 28.6, 22.6	18.4, 16.1, 14.2	(0.21, 0.32), (0.20, 0.29), (0.19, 0.27)
Device 4	2.3	3.7, 4.7, 5.4	67.7, 53.2, 43.6	23.5, 24.0, 22.8	(0.53, 0.46), (0.51, 0.46), (0.51, 0.45)

^{a)} Applied voltage required to reach a luminance of 1 cd m⁻². ^{b)} Values at 1000, 5000 and 10000 cd m⁻², respectively.

To elucidate the origin of the high performance of the Device 1, three reference devices (Devices 2, 3 and 4) with the same configuration as Device 1 were fabricated by using double neat films of FPPCA (20 nm)/Bepp₂ (10 nm) and a single doping film of Bepp₂:FPPCA (D) (1.5 wt%, 30 nm) and FPPCA (H):BZQPG (0.8 wt%, 30 nm) as the EML, respectively (see **Figure S2** in ESI file). The efficiency-luminance (PE/EQE-L) characteristics of Devices 2, 3 and 4 and their EL spectra at different luminances (1000, 2500 and 5000 cd m⁻²) are shown in **Figures 3c and 4**. The two emission peaks in Device 2 are assigned to: (i) direct exciton generation on the Bepp₂ and FPPCA molecules in the respective EML and (ii) singlet/triplet excited state energy in the blue EML transferring to the green EML; these are similar to those of Device 1 (**Figure 3b**) where both low-content dopants have almost no effect on the inherent properties of the host of Bepp₂ and FPPCA. Devices 3 and 4 comprise EMLs with